ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(a)** Both A and R are true and R is the correct explanation of A.	H^+ is needed for acidic behaviour.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene prevents contact with air/water.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
5	**(a)** 44 g	Mass is conserved: 12 + 32 = 44 g.
6	**(a)** 7	Complete neutralisation to NaCl.
7	**(d)** Y	Reacting with cold water indicates more reactivity.
8	**(a)** 44 g	1 mol CH_4 gives 1 mol CO_2.
9	**(b)** copper	CuO is reduced to Cu by hydrogen.
10	**(a)** plaster of Paris	It sets into gypsum.
11	**(c)** CuO and CO_2	Calcination: $CuCO_3 \longrightarrow CuO + CO_2$.
12	**(b)** ethene	C_2H_4 = 28.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Energy is needed to break bonds.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Heat could cause splashing of acid.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Oxide layer protects the metal.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Mass of CO_2 = 12 + 32 = 44 g (1); Law of conservation of mass (1).	1+1
17	CO_2 (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$, lime water turns milky (1).	1+1
18	Oxide is amphoteric (1); Zinc or aluminium (1).	1+1
19	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$; 60 g gives 176 g (1); 30 g gives 88 g CO_2 (1).	1+1
20	Gases NO_2 and O_2 escape (1); Only solid PbO remains; the total mass is still conserved (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Equation already balanced: $PbO + C \longrightarrow Pb + CO$ (1); PbO is the oxidising agent (1); C is the reducing agent (1).	1+1+1
22	pH 10 and 13 turn it pink (1); pH 1 and 4 neutralise a base (1); Acidity order: pH 1 > 4 > 7 > 10 > 13 (1).	1+1+1
23	ZnO is the oxidising agent and C is the reducing agent (1); Redox, reduction of metal oxide with carbon (1); Zinc is moderately reactive, so carbon can reduce its oxide (1).	1+1+1
24	Ratio C : H = 1 : 2, so empirical CH_2 (1); n = 42/14 = 3 (1); C_3H_6, propene (1).	1+1+1
	OR: Basic copper carbonate (1); Copper reacts with moisture, CO_2 and oxygen (1); Prevent by lacquer/paint or keeping dry (1).	1+1+1
25	X is $CaCO_3$, Y is CaO (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1+1
	OR: Salt water increases electrical conductivity (1); Faster electrochemical corrosion (1); Rust formation speeds up (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	A: no rust (1); B: no rust (1); C: rust forms (1); Conclusion: air (oxygen) and water are both needed (1); Prevention: galvanisation/painting (1).	1+1+1+1+1
	OR: HCl pH lower (1); HCl conducts better (1); Mg reacts faster with HCl (1); HCl is a strong acid ionising completely (1); CH_3COOH is weak, ionising partially (1).	1+1+1+1+1
27	Order: A > B > C > D (1); A: electrolysis of molten salt (very reactive) (1); D: occurs free or by heating its ore (1); Example set: Mg, Zn, Cu, Ag (1); Reason: a more reactive metal displaces a less reactive one (1).	1+1+1+1+1
	OR: Dehydration: conc. H_2SO_4, 443 K (1); Oxidation: alkaline $KMnO_4$ (1); $KMnO_4$ is oxidising agent (1); Esterification: conc. H_2SO_4 catalyst (1); Equations of each step (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Incomplete combustion gives soot.
2	**(a)** 16 g	40 - 24 = 16 g.
3	**(b)** 25 mL	Ratio 1 : 2.
4	**(b)** S	Each displaces the next, so S is lowest.
5	**(b)** 36 g	2 mol water = 36 g.
6	**(a)** zinc	Zn is more reactive; $ZnSO_4$ is colourless.
7	**(a)** $CuSO_4 \cdot 5\,H_2O$	Blue vitriol.
8	**(b)** copper	$CuO + H_2 \longrightarrow Cu + H_2O$.
9	**(a)** ethanoic acid	Vinegar.
10	**(c)** A is true but R is false.	A is true; R is false because nitrogen is unreactive.
11	**(a)** Both A and R are true and R is the correct explanation of A.	No free ions are produced.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Layers repel and crystal breaks.
13	**(c)** A is true but R is false.	A true; in graphite each carbon bonds to three others, so R is false.
14	**(d)** 36 g	4 + 32 = 36 g.
15	**(b)** 5	H^+ falls tenfold so pH rises by 1.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Plaster of Paris 145 g (1); Water lost 27 g (1).	1+1
17	Reduction of oxide is easier by carbon (1); So sulphides/carbonates are first converted to oxides (1).	1+1
18	$3n + 1 = 22$, n = 7 (1); C_7H_{16} (1).	1+1
19	$ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Carbon dioxide (turns lime water milky) (1).	1+1
20	Dilution is exothermic (1); Water floats and boils, spraying acid or cracking the glass (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	n-butane: 3 C-C, 10 C-H (1); isobutane: 3 C-C, 10 C-H (1); Same number of bonds; isomers (1).	1+1+1
22	Hydrogen gas (1); Burns with a pop sound near a flame (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1).	1+1+1
23	Acids react with copper/brass (1); Toxic salts form (1); Glass does not react with acids (1).	1+1+1
24	NaCl gives Na^+ and Cl^- ions (1); Sugar remains as neutral molecules (1); No ions in sugar solution (1).	1+1+1
	OR: C1-C4 gases (1); C5-C17 liquids (1); Higher are solids (wax) (1).	1+1+1
25	$Fe_3O_4 + 4\,H_2 \longrightarrow 3\,Fe + 4\,H_2O$ (2); H_2 is oxidised (1).	2+1
	OR: Add quick lime or slaked lime (1); Basic oxide neutralises soil acidity (1); Soil pH approaches neutral, improving crops (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Green to white (water lost), then reddish brown residue (1); $2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ (1); Gases: SO_2 and SO_3 (1); Smell of burning sulphur (1); Thermal decomposition (1).	1+1+1+1+1

	OR: Fixed H2O molecules per formula unit (1); $CuSO_4 \cdot 5\,H_2O$: blue to white (1); $FeSO_4 \cdot 7\,H_2O$: green to white then brown (1); $Na_2CO_3 \cdot 10\,H_2O$ loses water to give a white powder (1); Test: add water to the white residue - blue colour for $CuSO_4$ (1).	1+1+1+1+1
27	(a) Reacts with air and water (1); (b) Malleable, forms Al_2O_3 layer (1); (c) Pure gold is soft (1); (d) Like ions repel when layers shift (1); Each reason relates to property (1).	1+1+1+1+1
	OR: Butane and isobutane (1); But-1-ene and but-2-ene (1); Isomers: same formula, different structure (1); Homologues: differ by CH_2 (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $FeSO_4$ and $CuSO_4$	Zn is more reactive than Fe and Cu but less than Mg.
2	**(d)** 44.8 L	2 mol CO_2.
3	**(a)** Na_2CO_3	Carbonates release CO_2.
4	**(b)** $NaHCO_3$	Baking soda.
5	**(c)** copper	Cu and CuO.
6	**(a)** ethanol	Dehydration and oxidation.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Equal atoms mean equal mass.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Mild base neutralises acid.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Delocalised electrons carry the current.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Scum forms.
11	**(c)** 40 mL	Ratio H2 : O2 = 2 : 1.
12	**(b)** 3	pH = -log 10^{-3} = 3.
13	**(c)** Na_2O	Soluble basic oxides form alkalis.
14	**(a)** 88 g	1 mol gives 2 mol CO_2.
15	**(c)** increases slightly	Heavier Cu (63.5) replaces Fe (56) on the nail.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Protective Al_2O_3 film (1); Film prevents contact with air/water (1).	1+1
17	Propyne has a triple bond (1); 2 mol (1).	1+1
18	Iron (56) dissolves and copper (63.5) is deposited (1); Copper atoms are heavier, so the net mass rises (1).	1+1
19	pH becomes 5 (1); Still acidic (1).	1+1
20	Very exothermic, hydrogen catches fire (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	A, B more reactive than Fe (1); C is between Fe and Cu (1); Order: (A, B) > Fe > C > Cu > D (1).	1+1+1
22	$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (1); O_2 = 4 mol = 89.6 L (1); CO_2 = 2 mol = 44.8 L (1).	1+1+1
23	X is magnesium, Y is magnesium oxide (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Y is basic as $Mg(OH)_2$ forms with water (1).	1+1+1
24	Strong acid has lower pH (1); Weak acid ionises partially (1); Universal indicator shows red vs orange/yellow (1).	1+1+1
	OR: X is CuO, Y is copper (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1).	1+1+1
25	Graphite: each C bonded to 3, free electron (1); Diamond: each C bonded to 4, no free electrons (1); Fullerene: discrete C_{60} molecules (1).	1+1+1
	OR: Brown copper turns black: $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1); Black CuO turns brown again (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Zinc: bubbles of hydrogen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Copper: no reaction (1); Zinc is above hydrogen in the reactivity series (1); Copper is below hydrogen so cannot displace it (1).	1+1+1+1+1

	OR: First: neutral (pH 7) (1); Equal moles (1); Second: acid in excess (1); pH less than 7 (1); Excess H^+ = 0.0025 mol (1).	1+1+1+1+1
27	Purification of bauxite to Al_2O_3 (1); Molten alumina with cryolite (1); Cathode: $Al^{3+} + 3\,e^- \longrightarrow Al$ (1); Anode: $2\,O^{2-} \longrightarrow O_2 + 4\,e^-$; graphite burns (1); Cryolite lowers the temperature and raises conductivity (1).	1+1+1+1+1
	OR: 0.1 mol ethanol (1); $C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$ (1); CO_2 = 0.2 mol = 8.8 g (1); Water = 0.3 mol = 5.4 g (1); Volume = 4.48 L (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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